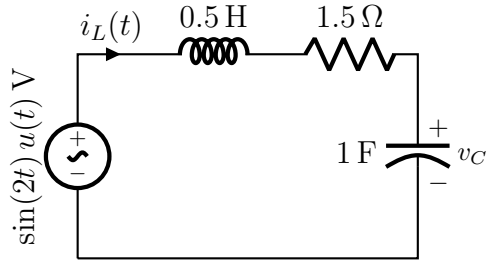
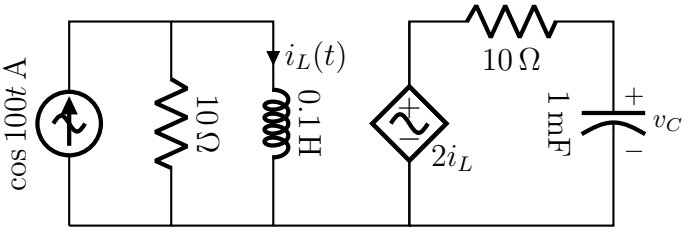
EE2015: Electric Circuits and Networks

Tutorial 8

(October 17, 2025)

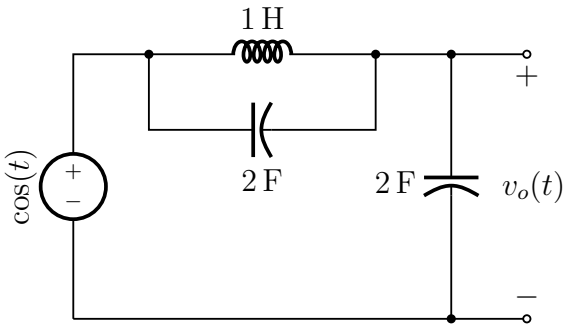
1. For the circuit shown on the right find $v_C(t)$ for $t \geq 0$. The initial conditions are: (i) $v_C(0^-) = 1\,\text{V}$, and (ii) $i_L(0^-) = 2\,\text{A}$.

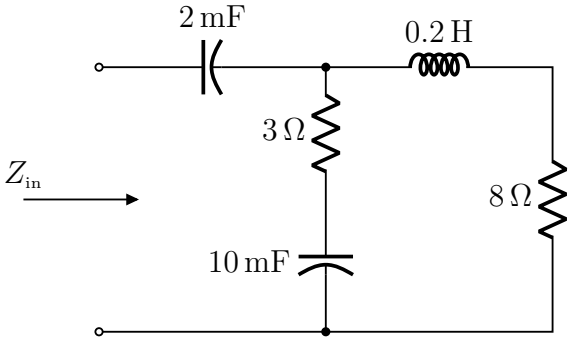


2. 

For the circuit shown on the left find $v_C(t)$.

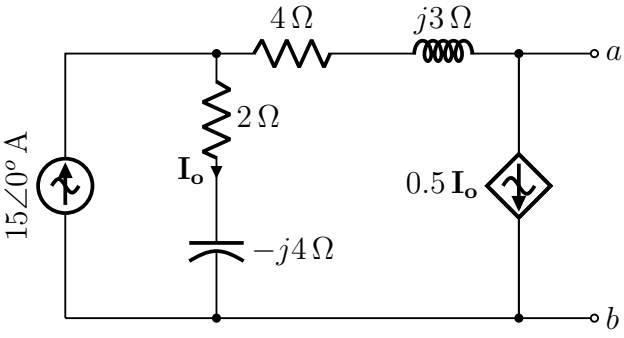
3. The circuit shown on the right is excited by $\cos(t)$. Find the simplified closed-form expression for $v_o(t)$.



4. 

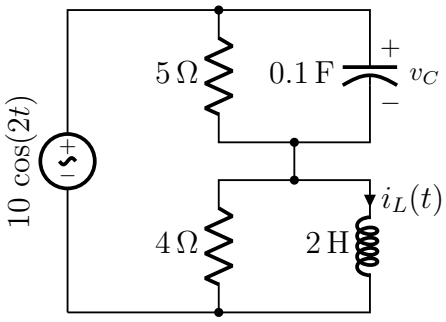
The circuit on the left operates at $\omega = 50\,\text{rad/s}$. Find the value of Z_{in}.

5. Find the Thevenin equivalent as seen from terminals ab for the circuit shown on the right.

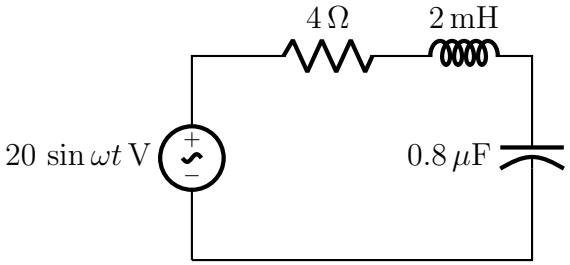


6.

Find $v_C(t)$ and $i_L(t)$ for the circuit show on the right.



7.



Consider the circuit shown on the left.

(a) Find the resonant frequency (ω_0) and the half-power frequencies (ω_1, ω_2).

(b) Calculate the Q-factor and bandwidth.

(c) Determine the amplitude of the current at ω_0, ω_1, and ω_2.